

src/app/changedFilesTree.ts

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T04:38:53.394Z

Overview

`src/app/changedFilesTree.ts` adds an in-memory tree for file-change data. It defines node types, path utilities, and functions to build, query, and flatten the tree for UI rendering.

Key changes

- **Interfaces & imports** – `DirNode`, `FileNode`, `TreeNode`, `FlatRow` and the imports `ChangeKind`, `FileChange` are added (R3, R10, R17, R19).
- **Path helpers** – `normalizePathSegments` (R34) normalises slashes and removes empty segments; `ancestorDirPaths` (R43) lists parent directories.
- **Tree construction** –
 - `ensureDir` (R64-R84) guarantees a directory node, deleting a conflicting file node (R73-R75).
 - `addFilePathToTree` (R87-R112) inserts a file path, creating intermediate directories.
 - `buildChangedFilesTreeRoot` (R114-R128) builds the root from a list of `FileChange` rows, sorting by depth first (R121-R124).
- **Flattening** – `flattenVisibleRows` (R131-R160) walks the tree, emitting a flat list of rows that respects an `expandedDirs` set.
- **Sorting** – `sortChildren` (R55-R62) orders children with directories first, then by name.

Impact

- **Deterministic ordering** – Children are sorted by type and name (R55-R62); rows are sorted by depth before insertion (R121-R124).
- **Linear flattening** – `flattenVisibleRows` visits each node once (R131-R160).
- **Self-contained** – No other modules are modified; the file is isolated.

Risks & follow-ups

- **Name collision** – `ensureDir` removes a file node if a directory with the same name is created (R73-R75). Verify that such collisions cannot occur in real data.
- **Path normalization** – `normalizePathSegments` replaces backslashes and collapses slashes (R34). Test on Windows paths to confirm consistency.
- **ExpandedDirs handling** – If `expandedDirs` is empty, all directories collapse (R131-R160). Confirm UI expectations.
- **Unused helper** – `ancestorDirPaths` (R43) is currently unused; document intended future use or remove to avoid dead code.